

Virgil Chan

Curriculum Vitae

CONTACT INFORMATION

Department of Mathematical Sciences
Indiana University–Purdue University Indi-
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RESEARCH INTERESTS

- Algebraic Topology:** Computational algebraic topology, Topological K-theory (especially topological K-theory for infinite CW-complexes), Homotopy Theory, Model Category Theory.
- Algebraic K-theory:** Computation of K-theories, trace methods in Algebraic K-theory, assembly maps in Algebraic K-theory, Farrell-Jones Conjecture.

EDUCATION

1. **Purdue University, West Lafayette**, West Lafayette, IN, United States
Doctor of Philosophy (Ph.D), [Mathematics](#), 2020 (Expected)
 - Advisor : Professor [Daniel A. Ramras](#) (IUPUI)
 - Dissertation Title : An Explicit Formula for the Loday Assembly (Tentative)
 - Area of Study : Algebraic Topology and Algebraic K-theory
2. **Indiana University–Purdue University Indianapolis**, Indianapolis, IN, United States
Graduate Student, [Mathematics](#), 2015–2020
3. **University of California, Davis**, Davis, CA, United States
 - (i). **Master of Arts (M.A.)**, [Mathematics](#), 2013–2015
 - (ii). **Bachelor of Science (B.S.)**, [Mathematics](#), 2010–2013
 - *Graduated with highest honours*
 - Thesis Title : *Topological K-Theory of Complex Projective Spaces*
 - Advisor : Professor Emeritus [Dmitry B. Fuchs](#)
 - Area of Study : Algebraic Topology

PROFESSIONAL

- 2018–present Co-organiser with Filipp Uvarov. [IUPUI Student-run Topological Quantum Field Theory Seminar](#).
 - Responsible for organising talks and sending out weekly announcements.
- 2016–present Co-organiser with Chenliang Huang, Kang Lu, Andrei Prokhorov, and Filipp Uvarov. [IUPUI Student-run Algebraic Geometry Seminar](#).
 - First student-run seminar at the IUPUI Math Department.
 - Responsible for organising talks and sending out weekly announcements.
 - Organised over 30 talks.

SERVICES	2016–2018	Secretary. IUPUI AMS Graduate Student Chapter . • During my term of service, the Graduate Student Chapter was featured in the Notices of AMS in May 2018 .	
HONOURS AND AWARDS	Graduate Student Travel Grants Highest Honours Distinction First-Year Scholar	American Mathematical Society University of California, Davis University of California, Davis	2019 Spring-2013 Fall-2010
CONFERENCE PRESENTATIONS	[1] An Explicit Formula for the Loday Assembly. In: AMS Special Session on Homotopy Theory and Algebraic K-theory at Binghamton University, 13-October-2019. [2] The Steinberg Group and Algebraic K-theory II: Quillen’s Algebraic K-theory. In: Modern Analysis and Geometry Seminar at Indiana University–Purdue University Indianapolis, 2-October-2018. [3] The Steinberg Group and Algebraic K-theory I: Milnor’s Algebraic K-theory. In: Modern Analysis and Geometry Seminar at Indiana University–Purdue University Indianapolis, 25-September-2018. [4] Introduction to Category Theory Through Algebra and Topology. In: IUPUI AMS Graduate Student Chapters Mini-course at Indiana University–Purdue University Indianapolis, 28-September, 3-October, 5-October-2017. • Three lectures given to graduate students. [5] The Assembly Map and the Isomorphism Conjecture in Algebraic K-theory. In: Modern Analysis and Geometry Seminar at Indiana University–Purdue University Indianapolis, 26-September-2017. [6] Introduction to Cohomology Operations. In: Student-Run Topology/Geometry Seminar at University of California, Davis, 19-May-2015. [7] Spectra and Cohomology Theories. In: Student-Run Topology/Geometry Seminar at University of California, Davis, 12-May-2015. [8] Groups, Spaces, and Bundles. In: Student-Run Topology/Geometry Seminar at University of California, Davis, 3-February-2015. [9] What is K-theory? In: Student-Run Topology/Geometry Seminar at University of California, Davis, 29-April-2014. [10] A Brief Introduction to Lower K-Theory. In: Student-Run Algebraic Geometry Seminar at University of California, Davis, 18-October-2012.		
RESEARCH PAPERS	[1] Topological K-Theory of Complex Projective Spaces. 2013. arXiv:1303.3959		
EXPOSITORY WRITINGS	[1] On the Adjunction Between Geometric Realisation and Singular Set. pdf [2] There Are Exactly Nine Model Structures on the Category of Sets, Modulo Axiom of Choice. pdf		

1. Indiana University–Purdue University Indianapolis¹, Indianapolis, IN **2015–present**

(i). Instructor

Fall 2019	MATH 11000	<i>Fundamentals of Algebra</i>
Spring 2019	MATH 11100	<i>Algebra</i>
Fall 2018	MATH 16500	<i>Analytic Geometry and Calculus I</i>
Spring 2018	MATH 22100	<i>Calculus for Technology I</i>
Fall 2017	MATH 11000	<i>Fundamentals of Algebra</i>

(ii). Teaching Assistant (instructor for recitation section)

Responsible for leading recitation sections, creating and grading exams, grading homework.

Spring 2017	MATH 16600	<i>Analytic Geometry and Calculus II</i>	for	Professor Jeffrey Watt
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(iii). Graduate Reader

September 2015–Present

Responsible for grading mathematics homework and quizzes.

Fall 2016	MATH 45300	<i>Beginning Abstract Algebra</i>
	MATH 55300	<i>Abstract Algebra</i>
Spring 2016	MATH S166	<i>Honors Analytic Geometry and Calculus II</i>
	MATH 27600	<i>Discrete Math</i>
	MATH 32101	<i>Elementary Topology</i>
	MATH 35100	<i>Elementary Linear Algebra</i>
Fall 2015	MATH S165	<i>Honors Analytic Geometry and Calculus I</i>
	MATH 22200	<i>Calculus for Technology II</i>
	MATH 45300	<i>Beginning Abstract Algebra</i>

2. University of California, Davis², Davis, CA

2013–2015

(i). Teaching Assistant

Responsible for leading discussion sections, grading exams. For some classes, I have also had the opportunity to write quizzes for students.

Spring-2015	MAT 108	<i>Introduction to Abstract Mathematics</i>	for	Dr. Jogia Bandyopadhyay
	MAT 22B	<i>Differential Equations</i>	for	Dr. David Cherney
Winter-2015	MAT 17B	<i>Calculus for Biology and Medicine</i>	for	Dr. Lawrence Marx
Fall-2014	MAT 108	<i>Introduction to Abstract Mathematics</i>	for	Dr. Michael Bishop

¹100-299 Lower Division Courses, 300-499 Upper Division Courses, 500-699 Graduate Level Courses.

²1-99 Lower Division Courses, 100-199 Upper Division Courses.

Spring-2014	MAT 21A	<i>[Differential] Calculus</i>	for	Dr. Vidya Venkateswaran
	MAT 108	<i>Introduction to Abstract Mathematics</i>	for	Dr. Eli Goldwyn
Winter-2014	MAT 17A	<i>Calculus for Biology and Medicine</i>	for	Dr. Sonya Berg
Fall-2013	MAT 21B	<i>[Integral] Calculus</i>	for	Professor Benjamin Morris

(ii). Graduate Reader

Responsible for grading mathematics homework in lower division courses.

Summer Session II-2014	MAT 21C	<i>[Sequential Series and Multivariable] Calculus</i>
	MAT 22A	<i>Linear Algebra</i>
Summer Session I-2014	MAT 22B	<i>Differential Equations</i>

CONFERENCES,
WORKSHOPS
ATTENDED

- [1] [Homotopy Theory Summer Berlin 2018](#). Zuse Institute Berlin and Technische Universität Berlin . 20–29-June-2018.
- [2] [2017 Midwest Topology Summer School on Trace Methods in Algebraic K-Theory](#). Indiana University Bloomington. 7–11-August-2017.
- [3] [Summer School on K-Theory and Related Fields](#). Hausdorff Research Institute for Mathematics. 19–23-June-2017.
- [4] AMS Special Session on Homotopy Theory. In: [American Mathematical Society Spring Central Sectional Meeting](#). Indiana University, Bloomington. 1–2-April-2017.
- [5] [Workshop: The Farrell-Jones Conjecture](#). Hausdorff Research Institute for Mathematics. 24–28-October-2016.
- [6] [Graduate Student Topology and Geometry Conference 2016](#). Indiana University, Bloomington . 2–3-April-2016.
- [7] [Geometry, Groups, and Dynamics Spring 2016](#). Indiana University–Purdue University Indianapolis. 5-March-2016.
- [8] [Wabash Modern Analysis Seminar](#). Wabash College. 27-February-2016.
- [9] [IU/PU/IUPUI Joint Topology Seminar](#). Indiana University–Purdue University Indianapolis. 5-December-2015, 16-April-2016, 12-November-2016.
- [10] [Annual Wabash Conferences at IUPUI](#). Indiana University–Purdue University Indianapolis. 12–13-September-2015.
- [11] [The Mathematics of Quantum Theory](#). University of California, Davis. 23–26-May-2014.
 - A conference focusing on recent developments in topological quantum field theory and topological strings and celebrating sixty years of spectacular contributions to mathematics and physics by Professor [Albert Schwarz](#).
- [12] [Introductory Workshop: Algebraic Topology](#). Mathematical Sciences Research Institute. 27-January-2014

- [13] West Coast Operator Algebra Seminar. [University of California, Davis](#). 26–27-October-2013

- VOLUNTEER WORK [1] [Annual Wabash Conferences at IUPUI](#). [Indiana University–Purdue University Indianapolis](#). 12–13-September-2015, 17–18 September-2016, 15–16 September-2018, 21–22 September-2019.
- Student helper.
- [2] Regional Mathcounts Competition. Sacramento Valley Chapter (03) MATHCOUNTS. [University of California, Davis](#). 7-February-2015.
- Grader
 - Responsible for checking students’ tests against the answer keys and filling out team scoring reports.
- [3] Math Café. [University of California, Davis](#). Spring-2014–Spring-2015.
- Tutor
 - Math Café is a mathematics-related club at UC Davis and was founded with the goal of encouraging students to succeed and continue their educations in mathematics fields.
- [4] [The Mathematics of Quantum Theory](#). [University of California, Davis](#). 23–26-May-2014.
- Student helper.
- [5] Galois Group Fundraiser. [University of California, Davis](#). 7-December-2013, 16-March-2014, 6-June-2014, 13-December-2014, 5-June-2015.
- Volunteer for tutoring services to help calculus students studying for their finals.

COMPUTER SKILLS

Mathematical Packages	:	Mathematica
Statistical Packages	:	R
Applications	:	L ^A T _E X, common Windows, Microsoft Office
Operating Systems	:	Linux, Windows.

REFERENCES AVAILABLE TO CONTACT

Dr. Daniel A. Ramras (Doctoral Advisor)
Associate Professor, [Mathematics](#)
[Indiana University–Purdue University Indianapolis](#)
✉: dramras@iupui.edu

Dr. John G. Miller
Associate Professor Emeritus, [Mathematics](#)
[Indiana University–Purdue University Indianapolis](#)
✉: jmiller@math.iupui.edu

Dr. Dmitry B. Fuchs (Undergrad Advisor)
Professor Emeritus, [Mathematics](#)
[University of California, Davis](#)
✉: fuchs@math.ucdavis.edu

Dr. Ronghui Ji
Associate Professor, [Mathematics](#)
[Indiana University–Purdue University Indianapolis](#)
✉: ronji@iupui.edu

Dr. Carl C. Cowen (Teaching Reference)
Professor, [Mathematics](#)
[Indiana University–Purdue University Indianapolis](#)
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Dr. Jerome Kaminker
Visiting Scholar, [Mathematics](#)
[University of California, Davis](#)
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